

Fundamentals of Modern VLSI Devices

SECOND EDITION

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Preface to the first edition

It has been fifty years since the invention of the bipolar transistor, more than forty years since the invention of the integrated-circuit (IC) technology, and more than thirty-five years since the invention of the MOSFET. During this time, there has been a tremendous and steady progress in the development of the IC technology with a rapid expansion of the IC industry. One distinct characteristic in the evolution of the IC technology is that the physical feature sizes of the transistors are reduced continually over time as the lithography technologies used to define these features become available. For almost thirty years now, the minimum lithography feature size used in IC manufacturing has been reduced at a rate of $0.7\times$ every three years. In 1997, the leading-edge IC products have a minimum feature size of $0.25\text{ }\mu\text{m}$.

The basic operating principles of large and small transistors are the same. However, the relative importance of the various device parameters and performance factors for transistors of the $1\text{-}\mu\text{m}$ and smaller generations is quite different from those for transistors of larger-dimension generations. For example, in the case of CMOS, the power-supply voltage was lowered from the standard 5 V, starting with the 0.6- to $0.8\text{-}\mu\text{m}$ generation. Since then CMOS power supply voltage has been lowered in steps once every few years as the device physical dimensions are reduced. At the same time, many physical phenomena, such as short-channel effect and velocity saturation, which are negligible in large-dimension MOSFETs, are becoming more and more important in determining the behavior of MOSFETs of deep-submicron dimensions. In the case of bipolar devices, breakdown voltage and base-widening effects are limiting their performance, and power dissipation is limiting their level of integration on a chip. Also, the advent of SiGe-base bipolar technology has extended the frequency capability of small-dimension bipolar transistors into the range previously reserved for GaAs and other compound-semiconductor devices.

The purpose of this book is to bring together the device fundamentals that govern the behavior of CMOS and bipolar transistors into a single text, with emphasis on those parameters and performance factors that are particularly important for VLSI (very-large-scale-integration) devices of deep-submicron dimensions. The book starts with a comprehensive review of the properties of the silicon material, and the basic physics of p-n junctions and MOS capacitors, as they relate to the fundamental principles of MOSFET and bipolar transistors. From there, the basic operation of MOSFET and bipolar devices, and their design and optimization for VLSI applications are developed. A great deal of the volume is devoted to in-depth discussions of the intricate interdependence and subtle tradeoffs of the various device parameters pertaining to circuit performance and manufacturability. The effects which are particularly important in small-dimension devices,